

Deep Learning Segmentation Approach Notes

Motivation: Dr. Zare has identified some issues with our existing segmentation implementation. While it seems to work great on some images, variations in lighting and other image characteristics can cause issues where not all of the pores are segmented correctly. To address this, we are looking into an approach that uses deep learning rather than traditional image processing to segment pore impurities.

Proposed Methods: The following is a breakdown of the methods we have researched. After the initial research phase, we need to implement and test each method. The ideal method should be both very fast and accurate, however there will probably be some trade off as it may be difficult or impossible to have both computational speed and accuracy optimized at the same time. The goal is to find a good balance that suits our needs.

Cellpose

Description:

Cellpose is a deep learning model designed for general-purpose cell segmentation. It was trained on over 70,000 images from diverse cellular microscopy sources, including SEM and fluorescence images, and has become a widely used tool in biomedical imaging. Unlike models that require domain-specific retraining, Cellpose is intended to generalize across many types of cell morphologies, imaging modalities, and staining methods.

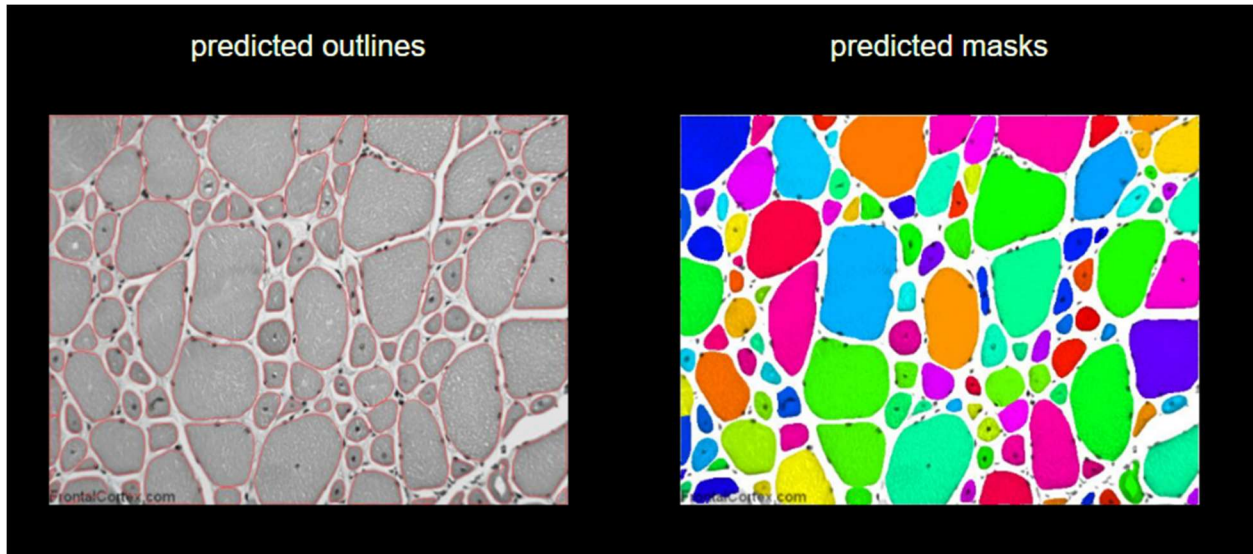
Pros:

- Strong performance on complex, irregular cellular shapes with varying orientations and light conditions.
- Out-of-the-box usability with no additional training needed for most biological datasets.
- Supports multiple cell types and modalities, making it highly versatile.
- Has an active research community and good documentation, meaning troubleshooting and extensions are accessible.

Cons:

- Optimization may be required if data diverges significantly from cellular morphologies (e.g., non-biological microscopy).

- Computationally expensive, especially when run on large datasets without GPU acceleration.
- While flexible, it is ultimately tuned for biological samples, so transferability to materials science or non-cell structures can be limited.



Segment Anything for Microscopy (SAM-Microscopy)

Description:

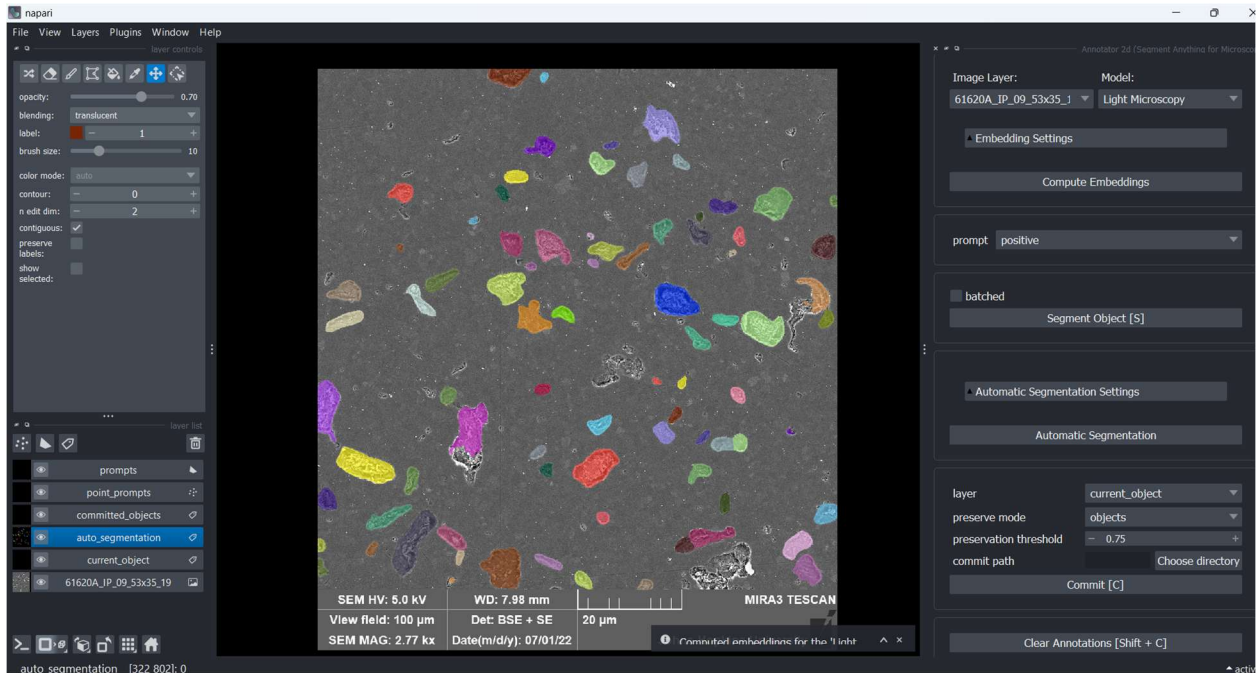
Segment Anything for Microscopy is a specialized, fine-tuned version of Meta's Segment Anything Model (SAM), adapted for microscopy and materials science tasks. Whereas SAM is designed for general-purpose segmentation across real-world photos, this version leverages microscopy-specific data to better detect microstructures such as grains, pores, and inclusions. It also provides a graphical user interface for annotation and further fine-tuning, making it more practical for scientific workflows.

Pros:

- Built on SAM's extremely powerful architecture but specialized for microscopy and materials science data.
- Improved accuracy on microstructures compared to base SAM.
- Integrated annotation GUI allows researchers to iteratively refine segmentation and retrain models.
- Can generalize to a wide range of microscopy techniques (SEM, TEM, optical).

Cons:

- High computational demand—requires powerful GPUs for smooth use.
- The annotation/tuning process can be time-intensive if significant customization is needed.
- May still underperform on niche or unusual materials datasets compared to a purpose-trained model.



NASA Pretrained Microscopy Models

Description:

Developed by NASA, this collection of pretrained models uses convolutional neural network (CNN) architectures trained on a large microscopy dataset of over 110,000 images. Unlike cell-focused models, this dataset includes materials science microscopy, making it more directly relevant to engineering and research applications outside of biology. The models also leverage MicroNet weights for improved transferability across microscopy tasks.

Pros:

- Directly trained on microscopy data (including materials), making it more relevant than purely biomedical models.
- Integrates natively with PyTorch for easy experimentation and fine-tuning.
- Open-source and already validated in aerospace and materials research contexts.

- Can be fine-tuned for specific datasets or applied directly out-of-the-box.

Cons:

- CNN-based architectures are less flexible than transformer-based approaches like SAM, potentially limiting generalization to highly novel datasets.
 - May underperform on extremely fine-grained or complex structures compared to newer models.
 - While pretrained on microscopy, performance still depends on how close the target dataset is to NASA's training data.
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SAM (Segment Anything Model)

Description:

SAM (Segment Anything Model), developed by Meta, is a cutting-edge segmentation model trained on one of the largest labeled image datasets ever created. Using a vision transformer (ViT) backbone, it was designed for “zero-shot” segmentation—able to adapt to new tasks and objects without additional training. SAM represents a paradigm shift in segmentation research, with applications ranging from natural images to scientific data.

Pros:

- Extremely powerful and flexible: designed to segment essentially any object.
- Large-scale training data and transformer backbone make it adaptable to many image domains.
- Performs well with limited user input (clicks, boxes, prompts).
- Strong community support and continual updates from Meta.

Cons:

- Trained primarily on natural and real-world images, not microscopy, so domain adaptation may be necessary.
- Very computationally expensive, often requiring high-end GPUs and significant memory.
- May produce overly general results on microscopic data without fine-tuning, since small-scale texture features differ from real-world data.